Summary Report:

Steps taken from API to Cleaning to Classification to Visualization:

I first connected to Reddit using the PRAW library and fetched 100 posts from two subreddits named datascience and gameofthrones. Each post had a title, body, author, upvotes, comments and time-stamp. Then I cleaned the text, removing links, punctuation and common words like "the" and "is." I split each post into individual words after cleaning and converted all of them to lower case. Then I used VADER to check if each post is positive, negative or neutral. I also trained a Naive Bayes model on the most common 200 words to predict what subreddit a post would belong to. Finally I loaded the data into Power BI and built charts to show the outcomes

Tools and Libraries Used:

All the code was written in Python. Used PRAW to scrape the posts and connect to reddit. The text was cleaned and stopwords were removed using NLTK. The sentiment of each post was determined using VADER from NLTK . Classifying posts into subreddits using Naive Bayes from NLTK. Pandas was used to save and handle data in tables. Charts were generated in Python using MatplotLib and Seaborn. Finally, the dashboard was developed in Power BI Desktop with interactive visuals and filters.

Challenges Faced During Integration:

The first challenge was setting up the Reddit API as I had to create an app on the Reddit website and get the client id and secret which was confusing at first. The second difficulty was that some posts had empty body text so I had to deal with that by replacing it with No body. Another problem was that VADER did not always produce accurate results for short Reddit titles, as they lack the necessary words to correctly detect sentiment. The fourth challenge was importing the word cloud visual into Power BI as it was not available by default and needed to be installed separately. The fifth challenge was that the majority of the posts were collected on the same date, therefore the time series line chart did not show much variation.

Future Improvements:

I would like to collect posts automatically every hour in the future so the data is always fresh and up to date. I would also like to use a better model like HuggingFace transformers for sentiment which gives more accurate results than VADER. I would also gather data from more subreddits to make the analysis more useful. Another improvement would be creating a live website using Streamlit where anyone can type a subreddit name and see the results immediately. Lastly I would like to be able to publish the Power BI dashboard online so it can be shared with others easily without them having to install Power BI.

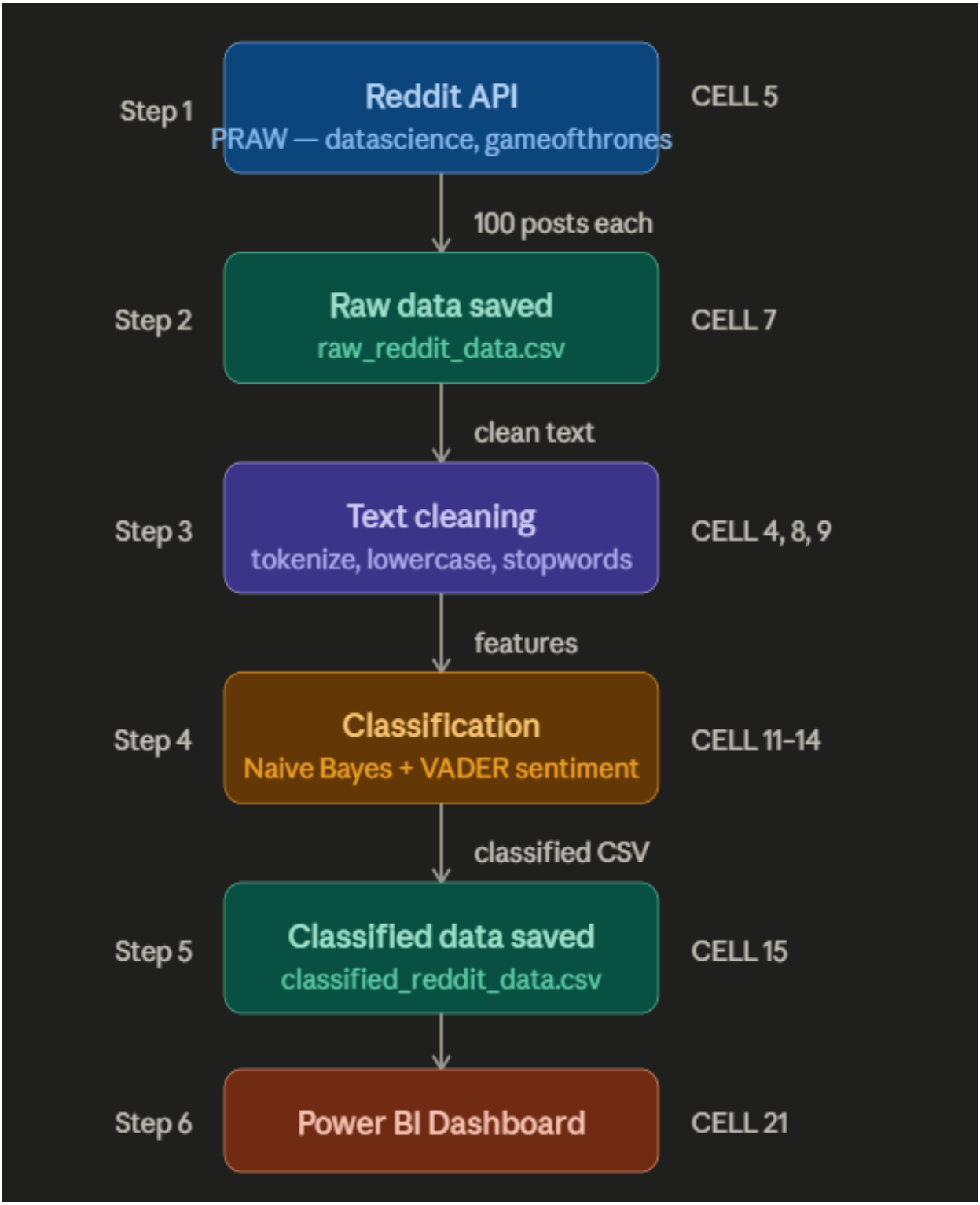
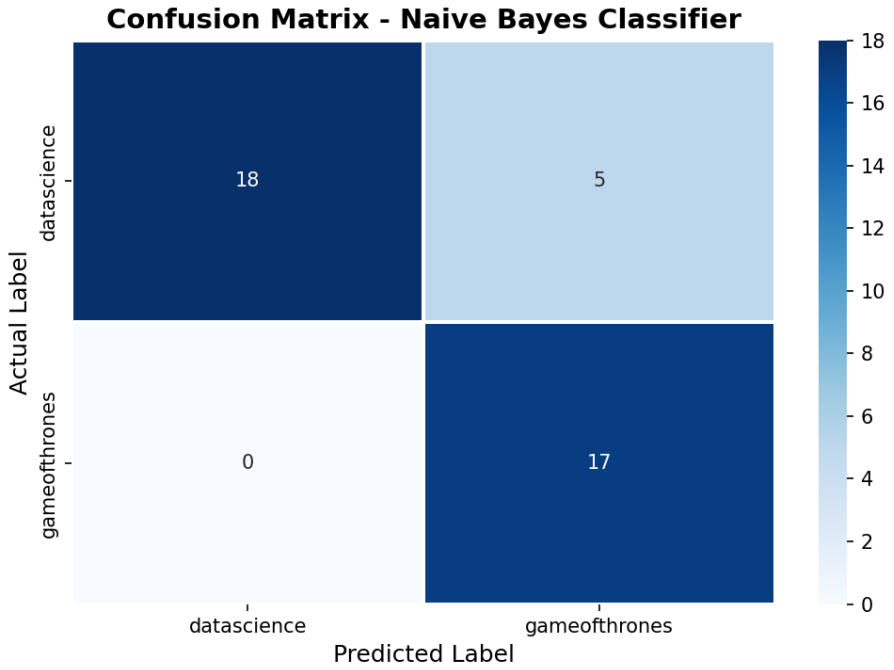
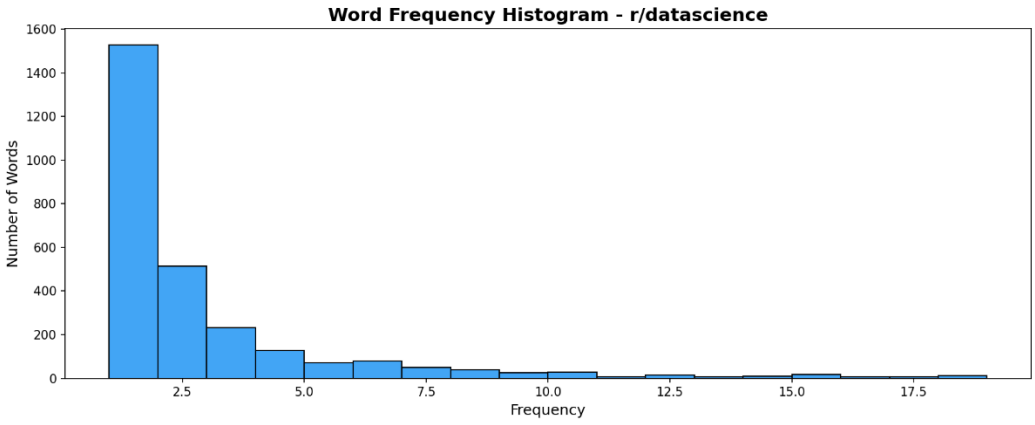


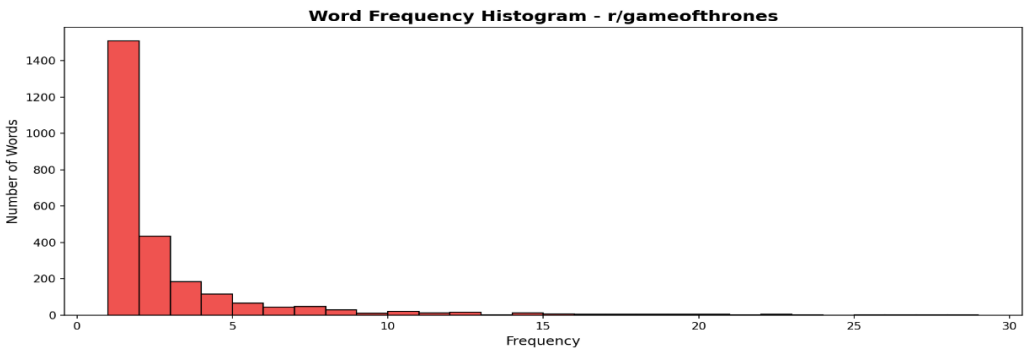
Figure 1 pipeline overview diagram

Confusion Matrix:

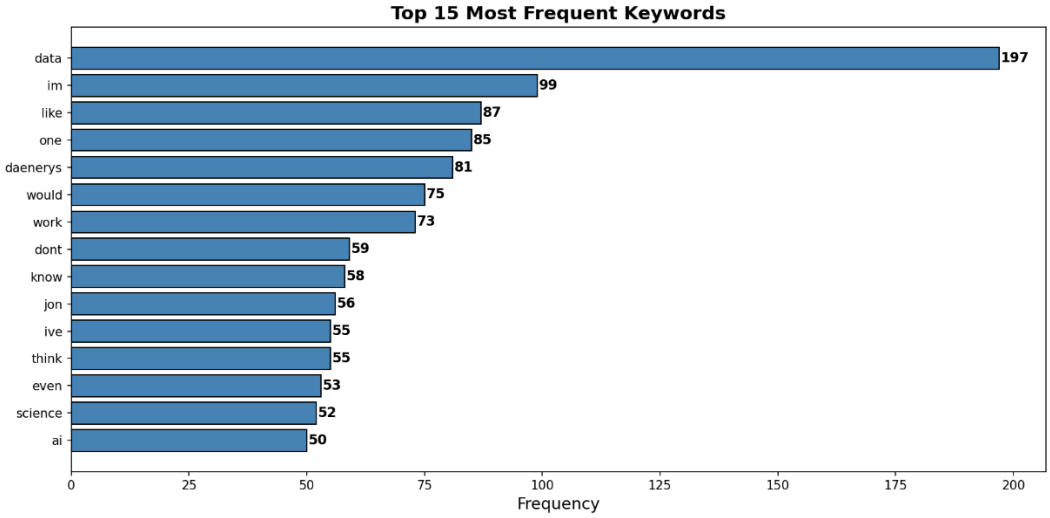


Histograms:

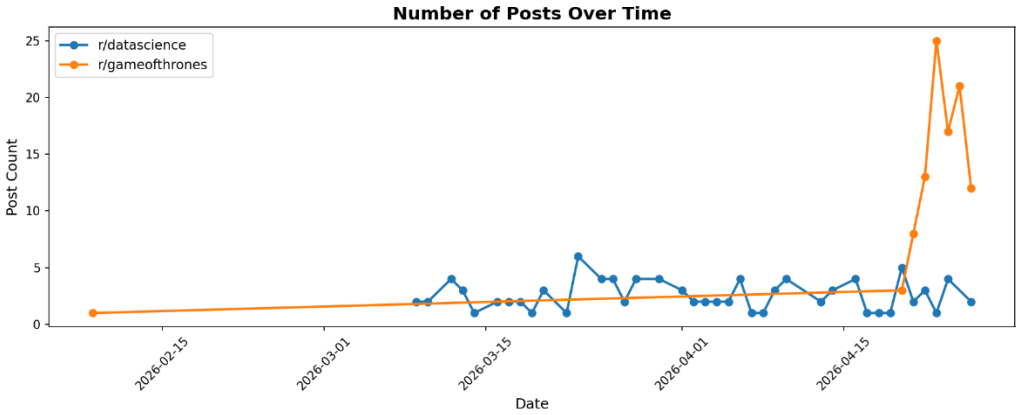




Frequent keywords:



Number of posts over time:



Sentiment Distribution:

